

ゲージ場形式による 量子誤り訂正符号の論理ゲート構成

Logical gates by gauge field formalism of quantum error correction

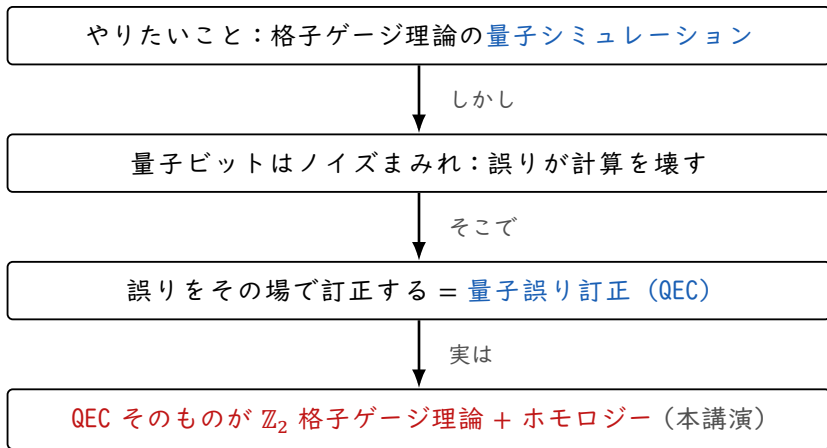


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日本物理学会 第81回年次大会 2026年9月17日 17aE532(13)

arXiv:2511.15224 (PTEP 2026, 073B12) に基づく

なぜ場の理論屋が量子誤り訂正？



準備：

量子誤り訂正と $\mathbb{Z}_2$ 格子ゲージ理論

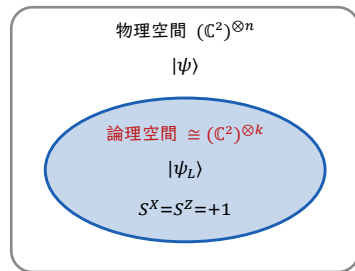
量子誤り訂正とは

- 論理情報 $\hookrightarrow$ 大きな Hilbert 空間
- 量子 (CSS) 符号：可換 X/Z スタビライザー

$$S^X = \prod_{j=1}^n (X_j)^{\gamma_j} \quad \text{or} \quad S^Z = \prod_{j=1}^n (Z_j)^{\tilde{\gamma}_j}$$

- 論理状態 $|\psi_L\rangle$ ：すべてのスタビライザーで固定

$$\forall S \quad S |\psi_L\rangle = +1 |\psi_L\rangle$$

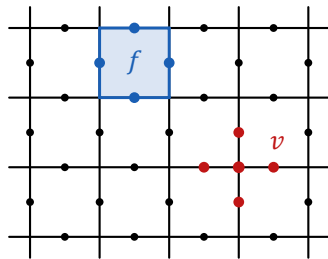


$\mathbb{Z}_2$ 格子ゲージ理論

- 実は量子符号は物理的には既知
- 2次元トーラス上の
Kogut-Susskind ハミルトニアン

$$H = -g^2 \sum_e X_e - \sum_p B_p - \sum_v A_v$$

- $B_p = \prod_{e \in \partial p} Z_e$ (磁束)
- $A_v = \prod_{e \ni v} X_e$ (Gauss 則)
- $Z_e =$ リンク変数 (ゲージ場)
- 運動項を落とす $\Rightarrow$ トポロジカル



辞書： $\mathbb{Z}_2$ 格子ゲージ理論 $\leftrightarrow$ QEC

- トポロジカル $\mathbb{Z}_2$ LGT = 量子符号

Gauss 則 A_v $\leftrightarrow$ X スタビライザー

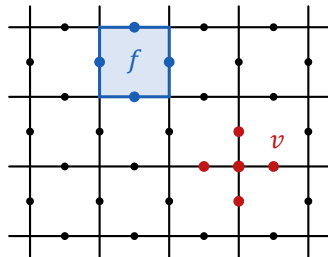
磁束 B_p $\leftrightarrow$ Z スタビライザー

リンク変数 Z_e $\leftrightarrow$ 量子ビット

- 基底状態 = 論理状態

$$A_v|\psi\rangle = B_p|\psi\rangle = +1|\psi\rangle$$

- スタビライザー（時空の多様体）の選択
= 量子符号の選択

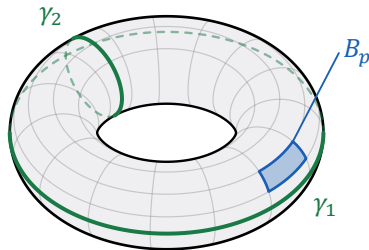


論理パウリ演算子 = Wilson ループ

- 論理パウリ演算子
 - すべてのスタビライザーと可換
 - スタビライザーの積ではない
- 論理状態への作用 = X/Z

$$\begin{aligned}\bar{Z}|0\rangle_L &= |0\rangle_L, & \bar{Z}|1\rangle_L &= -|1\rangle_L \\ \bar{X}|0\rangle_L &= |1\rangle_L, & \bar{X}|1\rangle_L &= |0\rangle_L\end{aligned}$$

- 論理パウリ = 非可縮ループ
⇒ ホモロジー／トポロジーで分類



パウリを超える論理ゲート

- Z : 位相反転、 X : ビット反転 $\Rightarrow$ 万能量子計算には不十分
- Solovay-Kitaev の定理 : クリフォード全体 + **非クリフォード** 1 つで十分
- 既知の手法 (トランスバーサル、格子手術など) は特定の符号族・幾何に依存
 $\Rightarrow$ 一般の符号に通用する**系統的な原理**は？

生成ゲート集合

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & \\ & i \end{pmatrix}, \quad CZ = \text{diag}(1, 1, 1, -1), \quad T = \begin{pmatrix} 1 & \\ & e^{i\pi/4} \end{pmatrix}$$

Problem

パウリを超える論理ゲートを一般の符号でどう構成するか？

本研究： ゲージ場による論理ゲート構成

Junichi Haruna
arXiv:2511.15224

ゲージ場 : 量子ビット → 射影子

- ゲージ場 = 量子ビットから演算子への線形写像

$$a(e_j) := P_j^- = |1\rangle\langle 1|_j = \frac{I - Z_j}{2}$$

$$a(e_i + e_j) := a(e_i) + a(e_j)$$

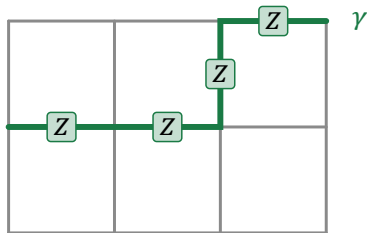
- 指数の方に乗せる = パウリ Z :

$$Z_j = \exp(i\pi a(e_j))$$

- パウリ Z の積 = Wilson 線

$$\prod_{i=1}^n (Z_i)^{r_i} = \exp(i\pi a(\gamma)) \leftrightarrow \exp\left(i\pi \int_{\gamma} a\right)$$

- 他の論理ゲートは？



論理演算子 = ゲージ不変：構成原理

- 論理演算子は符号空間を保つ： $S^\dagger U S \simeq U$ on $\mathcal{H}_c$
- X スタビライザーによる共役 = ゲージ変換

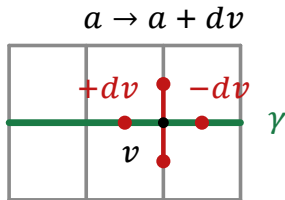
$$(S_v^X)^\dagger a S_v^X = a + dv \pmod{2}$$

- 論理 = ゲージ不変： $U(a(\gamma))$ の形なら O.K.

$$a(\gamma) = \oint_\gamma a \rightarrow \oint_\gamma (a + dv) = \oint_\gamma a$$

- どのゲートになる？ → 置換原理：物理 X, Z を論理 $\bar{X}, \bar{Z}$ に置換

$$\mathcal{O}(X, Z) \rightarrow \bar{\mathcal{O}}(\bar{X}, \bar{Z})$$



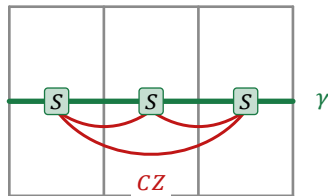
具体例：論理 S ゲート

- $S = \exp\left(i\frac{\pi I - Z}{2}\right) \Rightarrow \bar{S} = \exp\left(i\frac{\pi I - \bar{Z}}{2}\right)$
- $\bar{Z} = \exp(i\pi a(\gamma))$ をゲージ場の多項式に分解

$$\bar{S} = \exp\left(i\frac{\pi}{2}a(\gamma)^2\right)$$

- 物理ゲート分解

$$\bar{S}(\gamma) = \prod_{i \in \gamma} S_i \prod_{i < j \in \gamma} CZ_{ij}$$



γ 上：各量子ビットに S 、各ペアに CZ

まとめと展望

論理ゲート	ゲージ場表示
$\bar{S}(\gamma)$	$\exp(i\pi a(\gamma)^2/2)$
$\bar{H}(\gamma)$	$\bar{S}(\gamma) \exp(i\pi b(\tilde{\gamma})^2/2) \bar{S}(\gamma)$
$\overline{CZ}(\gamma_1, \gamma_2)$	$\exp(i\pi a(\gamma_1)a(\gamma_2))$
$\bar{T}(\gamma)$	$\exp(i\pi(2a^3 - 3a^2 + 2a)/4)$

- 😊 一般の CSS 符号に適用できる
- 😊 万能ゲート集合を含む
- 😊 論理作用はホモロジー類 $[\gamma]$ のみに依存
- 😬 誤り耐性はない（多体ゲートを含む）

ゲージ理論 ・ 量子誤り訂正 ・ ホモロジーは共通の数学的構造を持つ。

- 展望：誤り耐性をゲージ理論／ホモロジーの言葉で書く
非可換化 (quantum double 模型)、高次コホモロジー不変量と非クリフォードゲート

Thank you!

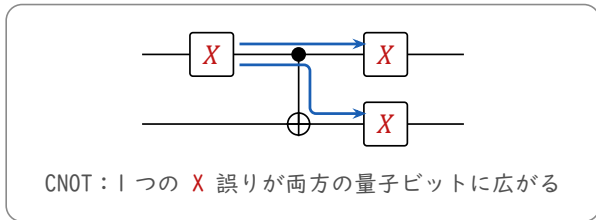
Backup

Homological origin of transversal gates

Junichi Haruna
arXiv:2602.14499

トランスバーサルゲートの必要性

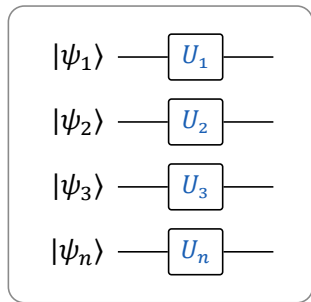
- 多量子ビットゲートは誤りを増幅する



- 誤り訂正能力を壊してしまう
- ではどう実装すべきか？ → 「トランスバーサルゲート」

トランスバーサルゲート

- 各量子ビットに独立な | 量子ビットユニタリ
- 誤りを増幅しない $\Rightarrow$ 誤り耐性
(fault-tolerant)
- 実現できるユニタリは限られる
 $\rightarrow$ Q. 万能? No



いつ・どのゲートがトランスバーサルか？

- **Eastin-Knill の定理** [Phys. Rev. Lett. 102, 110502 (2009)]
 - トランスバーサルゲートだけでは万能ゲート集合は作れない
 - ⇒ **何が**可能かの限界。与えられた符号で**どの**ゲートかは教えない
- 例は多いが**統一的な規則はない**

Problem 2

どの符号がどのゲートをトランスバーサルに実装できるか？

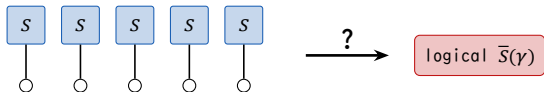
答えはホモロジー的障害で書ける (arXiv:2602.14499)

Transversal Z rotations

- Transversal S gate (as the simplest example):

$$U(\theta/2) := \prod_{j=1}^n \exp\left(i\pi \frac{\theta_j}{2} \frac{I - Z_j}{2}\right) = \prod_{j=1}^n (S_j)^{\theta_j}, \quad \theta_j \in \mathbb{Z}_4 = \{0, 1, 2, 3\}$$

- $S = \sqrt{Z}$ is the $\pi/2$ rotation, and $S^2 = Z$
- Goal: find θ such that $U(\theta/2)$ is **logical** and realizes $\sqrt{\bar{Z}}(\gamma)$
 - Existence **not guaranteed** (Problem 2 for S)

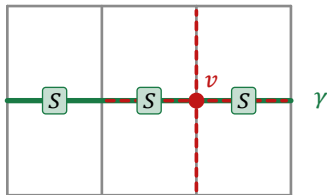


Transversal phase gates in LGT

- Transversal S = square root of Wilson loop:

$$\sqrt{\bar{Z}(\gamma)} = \exp\left(\frac{i\pi}{2} a(\gamma)\right)$$

- Gauge invariant? $\rightarrow$ No: sees $a(\gamma) \bmod 4$
- Fix $\rightarrow$ lift to $\mathbb{Z}_4$ field $\tilde{a}$: $\tilde{a} = a \bmod 2$
- Transversal S exists $\Leftrightarrow$ liftability



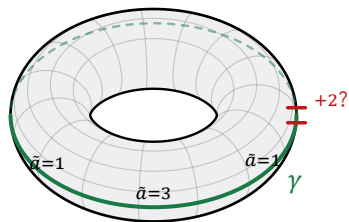
$a \rightarrow a + dv$: flips 2 links on γ

$$W(\gamma): \quad \times (-1)^2 = +1$$

$$\sqrt{W(\gamma)}: \quad \times i^2 = -1$$

Liftability is a topological question

- **Locally?** $\rightarrow$ always: $\tilde{a}(e) \in \{a(e), a(e) + 2\}$
- **Globally?** $\rightarrow$ can fail around nontrivial $[\gamma]$
- Decided by? $\rightarrow$ two **topological** data:
 - spacetime: $[\gamma] \in H_1$
 - gauge group: $\mathbb{Z}_2 \rightarrow \mathbb{Z}_4$
- Liftability $\rightarrow$ determined by **homology**



each link: lift exists

around $[\gamma]$: **+2 mismatch** possible

Main theorem

Theorem. Transversal implementability of logical S gate
 $\Leftrightarrow$ vanishing of two obstruction maps for $\bar{Z}(\gamma)$:

$$\beta_1([\gamma]) = \beta_2([\gamma]) = 0$$

- $\beta_1([\gamma]) := [(H_X \odot K)\gamma]$: additional condition mod 2
- $\beta_2([\gamma]) := [H_X(\gamma + H_Z^T \alpha)/2]$: liftability from $\mathbb{Z}_2$ to $\mathbb{Z}_4$

H_X = stack of X -checks, H_Z = stack of Z -checks, K = stack of $\{X\text{-checks}, \gamma\}$, $\odot$: row-wise product, $[\cdot]$ = class modulo trivial part

$\beta_2 =$ Bockstein obstruction

- $Z \rightarrow S$: halving the angle $\pi \rightarrow \pi/2 =$ **lifting the coefficients** $\mathbb{Z}_2 \rightarrow \mathbb{Z}_4$
- β_2 measures its obstruction
- **Transversal implementability = Existence of lift**

$$\begin{array}{ccccccc} & & [H\gamma/2] & & [\theta] & & [\gamma] \\ 0 & \longrightarrow & \mathbb{Z}_2 & \xrightarrow{\times 2} & \mathbb{Z}_4 & \xrightarrow{\text{mod } 2} & \mathbb{Z}_2 \longrightarrow 0 \\ & & \uparrow & & & & \uparrow \end{array}$$

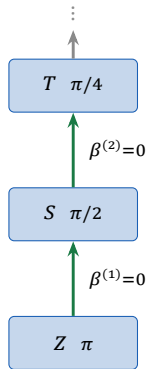
β_2 : Bockstein connecting map

keeping $[\gamma] = [\theta \bmod 2]$

Beyond Z and S

- Same argument for climbing the **Clifford hierarchy**: $Z \rightarrow S \rightarrow T \rightarrow \dots$
- Transversal $\Leftrightarrow$ a **ladder of vanishing obstructions** $\beta_1^{(v)} = \beta_2^{(v)} = 0$
- Gauge-invariant root $\exp(i\pi/2^{m-1} \phi \tilde{a})$
= **liftability** to $\mathbb{Z}_{2^m}$

$\Rightarrow$ Answer to Problem 2

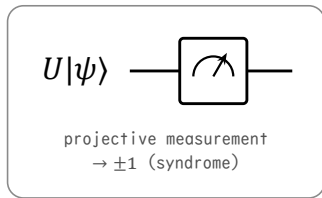


How to detect and correct?

- Measurement of stabilizers $\Rightarrow$ Error detection

$$|\psi_L\rangle \xrightarrow{\text{error}} U|\psi_L\rangle \xrightarrow{\text{measure}} S = \pm 1$$

- Flipped result (= “syndrome”)
 $\Rightarrow$ Error occurred!
- Error-correcting capability
= Min. weight of errors that
do not flip syndromes



Two problems, one dictionary

Problem 1

How to **construct** logical gates beyond Pauli in general?

⇒ Main Part 1: a gauge-field engine arXiv:2511.15224

Problem 2

Which code implements which gate **transversally**?

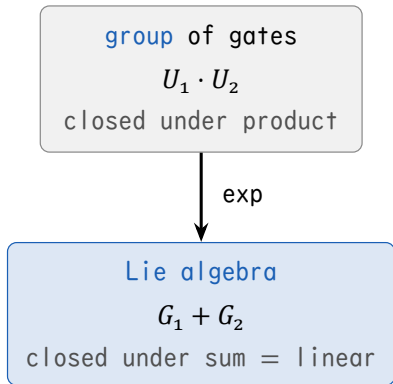
⇒ Main Part 2: a homological obstruction arXiv:2602.14499

Work in the exponent

- Gates form a **group**: closed under the product $U_1 U_2$ (nonlinear)
- Idea: push everything into the exponent

$$U = \exp(i G)$$

- Then generators just *add*: $G_1 + G_2$
⇒ a **linear space**, much simpler
- **Gauge field** = the generator G : an operator-valued cochain
linear map: qubits $\rightarrow$ operators



Why projectors?

- Every diagonal gate = one master formula:

$$U_{\text{diag}} = \exp\left(i f(P_1^-, \dots, P_n^-) \right)$$

- choosing the gate = choosing f :

$$Z = \exp(i\pi P^-)$$

same P^- ,

$$S = \exp\left(\frac{i\pi}{2} P^-\right)$$

only the angle changes

$$T = \exp\left(\frac{i\pi}{4} P^-\right)$$

$$CZ_{ij} = \exp\left(i\pi P_i^- P_j^-\right)$$

product $\rightarrow$ entangling

Logicality from the commutation relation

- Logicality $\Leftrightarrow$ commutativity on $\mathcal{H}_c$
- Commutation relation with X -stabilizer:

$$S^X(h)^\dagger U^\dagger\left(\frac{\theta}{2}\right) S^X(h) U\left(\frac{\theta}{2}\right) \\ = \underbrace{e^{i\pi\langle\theta,h\rangle/2}}_{\text{const. phase}} \underbrace{U(\theta \circ h)}_{\text{Z-string}}$$

$\theta \circ h$: entrywise product

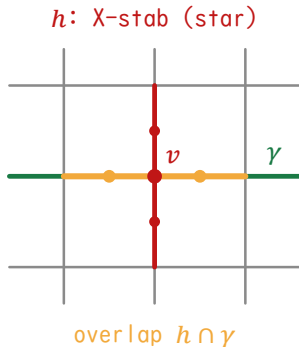
- Z -string must coincide with a stabilizer (and phase trivial)

$$\begin{array}{ccc} |\psi_L\rangle & \xrightarrow{S_v^X \simeq I} & |\psi_L\rangle \\ \downarrow U & & \downarrow U \\ U|\psi_L\rangle & \xrightarrow{S_v^X \simeq I} & US_v^X|\psi_L\rangle \\ & & \simeq S_v^X U|\psi_L\rangle \end{array}$$

$$S^\dagger U^\dagger S U \simeq I \quad \text{on } \mathcal{H}_c$$

Two logicality conditions on θ

- Two conditions on θ , **different moduli**:
 - Phase factor**: $H\theta = 0 \pmod{4}$
 - Z-string = stabilizer** ($\perp h, \gamma$):
 $(H \odot K)\theta = 0 \pmod{2}$
- Which logical Z ? $\rightarrow$ further ($S^2 = Z$) needed



H = stack of $\{h\}$, K = stack of $\{h, \gamma\}$; $\odot$: row-wise product

Simplifying $\bar{S}(\gamma)$ modulo 4

- $\bar{S}(\gamma)$ as a polynomial of gauge fields:

$$\bar{S}(\gamma) = \exp\left(\frac{i\pi}{2} \left(\frac{10}{3}a(\gamma) - 3a(\gamma)^2 + \frac{2}{3}a(\gamma)^3\right)\right)$$

- Simplified modulo 4 (since $S^4 = I$):

$$\bar{S}(\gamma) = \exp\left(\frac{i\pi}{2}a(\gamma)^2\right)$$

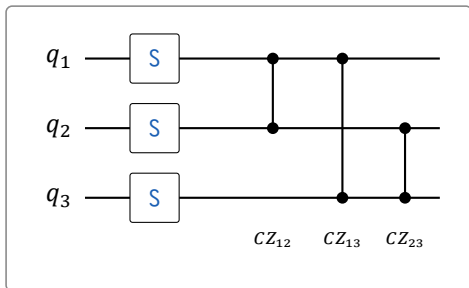
Physical gate decomposition ($n = 3$)

- Expansion of gauge fields polynomial

$$\begin{aligned}\bar{S}(\gamma) &= \exp\left(\frac{i\pi}{2}(z_1 + z_2 + z_3)^2\right) \\ &= \exp\left(\frac{i\pi}{2}(z_1 + z_2 + z_3)\right) \\ &\quad \times \exp(i\pi(z_1z_2 + z_1z_3 + z_2z_3))\end{aligned}$$

- First factor = S on each qubit,
second factor = CZ on every pair

$$\bar{S}(\gamma) = S_1 S_2 S_3 \cdot CZ_{12} CZ_{13} CZ_{23}$$



Summary of Main Part I

- **Dictionary:** $\mathbb{Z}_2$ lattice gauge theory $\rightleftharpoons$ quantum codes
- **Systematic construction:**
 - Logical gate = exponential of a polynomial of gauge fields
 - Replacement principle: $Z \rightarrow \bar{Z}$, $X \rightarrow \bar{X}$
 - Applicable to general quantum (CSS) codes

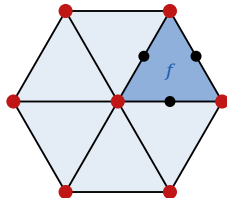
These gates are **neither transversal, nor fault-tolerant** in general.

When is a logical gate transversal? $\rightarrow$ Main Part 2

CSS codes = chain complexes over $\mathbb{Z}_2$

$$C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0$$

- Latticize a space into cells
- Chain complex: ∂ = boundary map
- Qubits on C_1 ; $H_X = \partial_1$, $H_Z^T = \partial_2$
- CSS condition = $\partial^2 = 0$



face = C_2 , edge/qubit = C_1 , vertex = C_0

Chain complexes across QEC

- Measurement error $\Rightarrow$ longer chain complexes

- Fault-complex

[Hillmann et al. 2025]

- Single-shot QEC

[Campbell 2018; Jacob-McLauchlan-Browne 2508.08191]

- Algebraic operation $\Rightarrow$ new family of QEC codes

- Hypergraph product codes $\rightarrow$ tensor product

[Tillich, Zémor 2014]

- Fiber-bundle codes $\rightarrow$ fiber bundle

[Hastings, Haah, O'Donnell 2021]

- Lifted product codes $\rightarrow$ lifted product

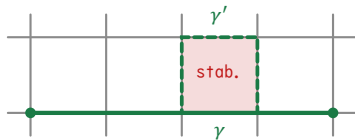
[Panteleev, Kalachev 2021]

- Code surgery $\rightarrow$ mapping cone

[Ide et al. 2025]

β_1, β_2 as homology maps

- So far: β from *one fixed* γ
- But: same logical $\bar{Z}$ for topologically equivalent loops
- Modding out by stabilizers
 $\Rightarrow \beta_1, \beta_2$ depend **only on topological equivalence class** $[\gamma]$



$$\gamma' = \gamma + \partial(\text{stab.})$$

$$[\gamma'] = [\gamma] \Rightarrow \beta(\gamma') = \beta(\gamma)$$

Steane $[[7, 1, 3]]$: transversal S yes, T no

- Self-dual CSS ($H_X = H_Z = H$, the $[7, 4, 3]$ Hamming code): a single logical class

gate	lift	obstruction	transversal?
$S(\pi/2)$	$1 \rightarrow 2$	$\beta_1^{(1)} = \beta_2^{(1)} = 0$	yes
$T(\pi/4)$	$2 \rightarrow 3$	$\beta_2^{(2)} \neq 0$	no

- S : lift $\hat{\theta} = 3 \cdot 1_7$ gives $\langle \hat{\theta}, 1_7 \rangle = 1 \pmod{4}$, so $\prod_i S_i^\dagger$ is logical S
- T : every logical $\theta^{(3)} = 2c_0 1_7 + 4 \sum_i c_i h_i$ is even, so $\langle \theta^{(3)}, 1_7 \rangle \neq 1 \pmod{8}$: no transversal T , even for non-uniform $\theta \in \mathbb{Z}_8^7$
- $[[15, 1, 3]]$ Reed-Muller: transversal T (backup).

The obstruction detects the absence of transversal T : Eastin-Knill, from obstruction theory.

Relation to high-energy physics?

- Transversal Pauli Z rotation gate at level $m = “2^{m-1}$ -root of Wilson loop”
- Transversal implementability = $\exists$ gauge-invariant 2^{m-1} -root of $W(\gamma)$
- Example: $m = 2 =$ transversal S (our running case)

$$\begin{array}{ccc} \text{gauge field side} & & \text{QEC operator side} \\ W(\gamma) = \exp\left(\oint_{\gamma} a\right) & \leftrightarrow & Z(\gamma) = \prod_i (Z_i)^{\gamma_i} \\ \\ W^{(2)}(\gamma^{(2)}) = \exp\left(\frac{1}{2} \oint_{\gamma^{(2)}} \tilde{a}\right) & \leftrightarrow & S(\gamma^{(2)}) = \prod_{i=1}^n (S_i)^{\gamma_i^{(2)}} \\ \text{with } W(\gamma) = (W^{(2)}(\gamma^{(2)}))^2 & & \end{array}$$

Our transversal S is the case $m = 2$: the **square root** of the Wilson loop.
 $m = 3$ (4th root) = transversal T on $[[15, 1, 3]] \rightarrow$ next.

[[15, 1, 3]] Reed-Muller: setup

- Setup: the [[15,1,3]] punctured Reed-Muller code. Physical $T^\dagger$ on every qubit $\rightarrow$ logical T .
- In this framework: uniform angle $\theta_i \equiv 1$, level $m = 3$, $\mathbb{Z}_8$ coefficients.
- **Tri-orthogonality** (every triple overlap of H^Z rows is even) $\Rightarrow \beta_1^{(\nu)}, \beta_2^{(\nu)}$ **vanish** for $\nu = 1, 2, 3$.

[[15, 1, 3]] Reed-Muller: transversal T

- $\Rightarrow$ A $\pi/4$ rotation of logical Z is transversal = transversal T .
- HEP reading: $m = 3$, so logical $\bar{T}$ is the $2^{m-1} = 4$ -th root of the Wilson loop:

$$\bar{T}(\gamma) = W(\gamma)^{1/4}$$

Textbook “tri-orthogonal $\Rightarrow$ transversal T ” = both obstructions vanish at uniform angle, level 3.

Recovering known transversality criteria

Known criterion	In this framework
Divisibility: a 2^m -divisible CSS code admits transversal $\exp(i\pi Z/2^m)$	$\beta_2^{(v)} = 0$ for uniform θ_v (the homology class lifts)
Tri-orthogonality $\Rightarrow$ transversal T	level $m = 3$ with $\beta_1^{(v)} = \beta_2^{(v)} = 0$ (uniform angle)
Quasi- / multi-level generalizations	non-uniform θ_v : the full classification

Divisibility and tri-orthogonality are the uniform-angle corner of a full homological classification (non-uniform angles give strictly more transversal gates).

Relation to Webster-Quintavalle-Bartlett

- Closest prior work [arXiv:2303.15615]: transversal diagonal gates from a **level structure** on the code
- At each level, the **two classifications coincide**
- Difference: closed-form / homological (here) vs algorithmic (there)

Same classification, complementary content; β_2 is the Bockstein-type refinement.

Two circuits for one $\bar{S}$

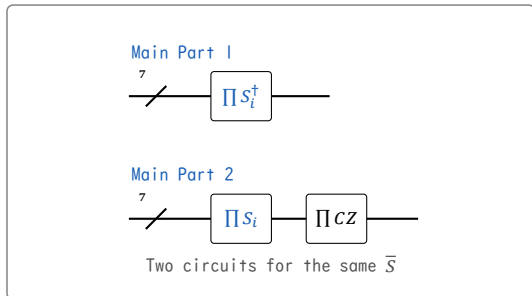
- Main Part 1 says

$$\bar{S} = \prod_{i=1}^7 S_i^\dagger$$

- Main Part 2 says

$$\bar{S} = \prod_{i=1}^7 S_i \cdot \prod_{1 \leq i < j \leq 7} CZ_{ij}$$

- Contradiction? What happens?



Resolved by a logical identity

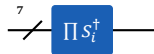
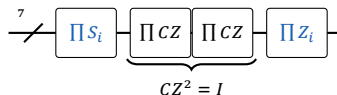
- CZ part + logical Z = logical identity

$$\prod_{1 \leq i < j \leq 7} CZ_{ij} \cdot \bar{Z} \simeq I$$

- Deform by a logical identity:

$$\prod_i S_i \cdot \prod_{i < j} CZ_{ij} \simeq \prod_i S_i \cdot \prod_i Z_i = \prod_i S_i^\dagger$$

- “Nice” logical gate $\Leftrightarrow$ “nice” logical identity



Deform $\bar{S}$ by a logical identity
 $\Rightarrow$ transversal $S^\dagger$

Relation to gauge-covariant codes

- Active line in this community: [encode a gauge theory as a code](#)
 - Gauss law at each vertex $\rightarrow$ a stabilizer [Roggero et al. 2024; Yao 2025]
 - Goal: fault-tolerant simulation of the gauge theory itself
- This work: the **opposite arrow**
 - Use gauge / homological structure to classify transversal gates of [any](#) CSS code

Same dictionary, opposite direction:
[gauge theory](#) $\rightarrow$ [code](#) (simulation) vs [code](#) $\rightarrow$ [gauge structure](#) (fault-tolerant logic).

Non-Abelian code and quantum double model

- Non-Abelian gauge groups $\rightarrow$ non-Abelian anyons = key to topological QC
- Quantum double model (Kitaev 1997):
 - A lattice model with the quantum double of a finite group G
 - $G = \mathbb{Z}_2$: reduces to the toric code
 - G is non-Abelian: realizes non-Abelian anyons (e.g. $G = S_3, D_4$)